Middle East Technical University
Department of Electrical and Electronics Engineering
EE301: Signals and Systems I

Solutions for Homework 1
October 18, 2017
If you face any problem or mistake please contact Ömer Çayır, *ocayir@metu.edu.tr*, DZ-10.

1.	(a)	$\underbrace{3\exp\left(j\frac{\pi}{2}\right)}_{3j}+\underbrace{3\exp\left(-j\frac{\pi}{2}\right)}_{-3j}+\underbrace{\exp\left(j\pi\right)}_{-1}-\underbrace{4\exp\left(-j\frac{\pi}{2}\right)}_{-4j}+4j+\underbrace{2\exp\left(j\frac{\pi}{6}\right)}_{\sqrt{3}+j}=(\sqrt{3}-1)+9j$
	(b)	$\frac{(1-j)(5-5j)(\sqrt{3}-j)}{10j\left(5-j5\sqrt{3}\right)}=\frac{\sqrt{2}\exp\left(-j\frac{\pi}{4}\right)5\sqrt{2}\exp\left(-j\frac{\pi}{4}\right)2\exp\left(-j\frac{\pi}{6}\right)}{10\exp\left(j\frac{\pi}{2}\right)10\exp\left(-j\frac{\pi}{3}\right)}$ $=\frac{1}{5}\exp\left(-j\frac{5\pi}{6}\right)$ $=\frac{1}{10}\left(-\sqrt{3}-j\right)$
	(c)	$\left(\frac{1}{\sqrt{2}}-\frac{1}{\sqrt{2}}j\right)^5=\left(\exp\left(-j\frac{\pi}{4}\right)\right)^5=\exp\left(-j\frac{5\pi}{4}\right)=\frac{-1}{\sqrt{2}}+\frac{1}{\sqrt{2}}j$
	(d)	$\left(\cos\frac{\pi}{6}-j\sin\frac{\pi}{6}\right)^6=\left(\exp\left(-j\frac{\pi}{6}\right)\right)^6=\exp\left(-j\pi\right)=-1$

2. We assume that

$$\exp(z)=\sum_{n=0}^{\infty}\frac{z^n}{n!}$$

for any $z\in\mathbb{C}$. Let $z=j\theta$,

$$\exp\left(j\theta\right)=\sum_{n=0}^{\infty}\frac{(j\theta)^n}{n!}=1+j\theta+\frac{(j\theta)^2}{2!}+\frac{(j\theta)^3}{3!}+\frac{(j\theta)^4}{4!}+\cdots=1+j\theta-\frac{\theta^2}{2!}-j\frac{\theta^3}{3!}+\frac{\theta^4}{4!}+\cdots$$

It is clear that even and odd n values correspond to real and imaginary parts, respectively.

$$\exp(j\theta)=\sum_{n=0}^{\infty}\frac{(-1)^n}{(2n)!}\theta^{2n}+j\sum_{n=0}^{\infty}\frac{(-1)^n}{(2n+1)!}\theta^{2n+1}$$

For cosine and sine functions, the Maclaurin series are known as

$$\cos\theta=\sum_{n=0}^{\infty}\frac{(-1)^n}{(2n)!}\theta^{2n}\qquad\sin\theta=\sum_{n=0}^{\infty}\frac{(-1)^n}{(2n+1)!}\theta^{2n+1}$$

By substituting them, we obtain

$$\exp\left(j\theta\right)=\cos\theta+j\sin\theta$$

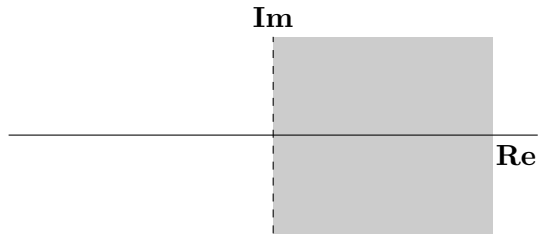
3. For $z = x + jy$,

$$\exp(z) = \exp(x + jy) = \exp(x)\exp(jy)$$

Using Euler's formula, we obtain

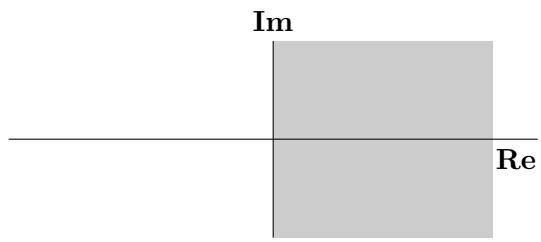
$$\exp(z) = \exp(x)[\cos(y) + j\sin(y)]$$

4. (a) On the complex plane, **Re** is the real axis and **Im** is the imaginary axis.



$$S_1 = \left\{ z \middle| \mathbf{Re}(z) > 0 \right\}$$

(b)

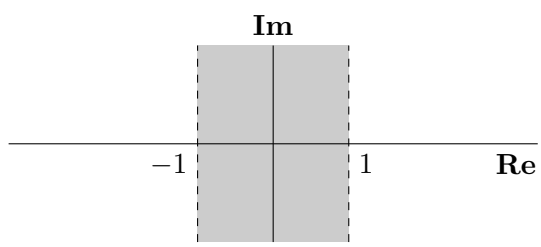


$$S_2 = \left\{ z \middle| \mathbf{Re}(z) \geqslant 0 \right\}$$

i. $S_1 \subset S_2$, and S_2 includes the imaginary axis, $\mathbf{Re}(z) = 0$.

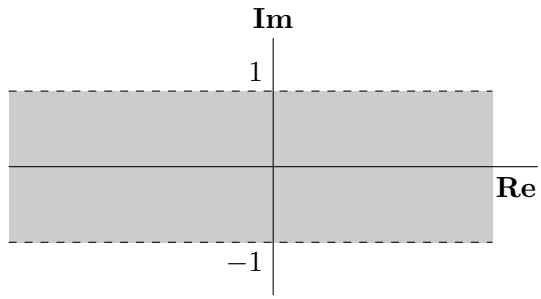
ii. $S_{im} = \{z = x + jy | x = 0\}$ is the set of complex numbers on the imaginary axis, and $S_2 = S_1 \cup S_{im}$. We can also relate them as $S_1 = S_2 \setminus S_{im}$ and $S_{im} = S_2 \setminus S_1$.

(c)



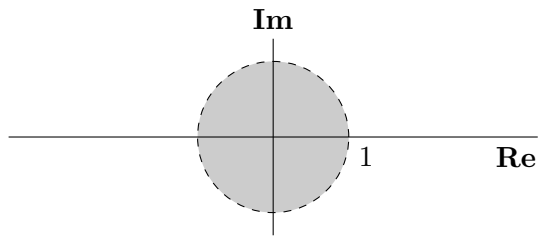
$$S_3 = \left\{ z \middle| -1 < \mathbf{Re}(z) < 1 \right\}$$

(d)



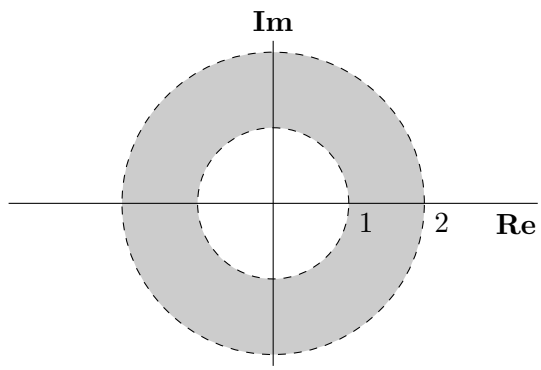
$$S_4 = \left\{ z \middle| -1 < \mathbf{Im}(z) < 1 \right\}$$

(e)



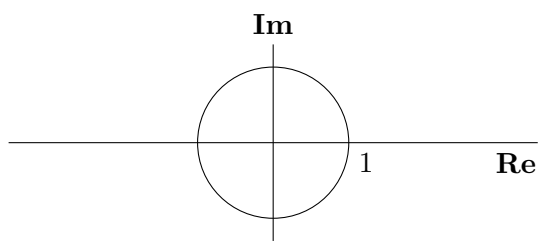
$$S_5 = \left\{ z \middle| |z| < 1 \right\}$$

(f)



$$S_6 = \left\{ z \middle| 1 < |z| < 2 \right\}$$

(g)



$$S_7 = \left\{ z \middle| |z| = 1 \right\}$$

The set S_7 is called the unit circle since it includes the circle of unit radius, i.e., of radius 1.

i. $S_{ud} = \left\{ z \middle| |z| \leqslant 1 \right\}$ is the set of all complex numbers z with modulus less than or equal to 1, the closed unit disk.

ii. $S_8 = \left\{ z \middle| 0 < |z| \leqslant 1 \right\}$ is the set of all complex numbers z with modulus less than or equal to 1 except $|z| = 0$. The set S_8 is called the punctured (closed) unit disk since it includes the unit circle (disk boundary) and a disk with radius 1 except the center of disk (the hole at the center).

5. (a) By using Euler's formula, we obtain the Cartesian form

$$(r\exp(j\theta))^n = (r\cos\theta + jr\sin\theta)^n$$

Then, using de Moivre's formula,

$$(r\cos\theta + jr\sin\theta)^n = r^n(\cos n\theta + j\sin n\theta)$$

we can show that

$$(r\exp(j\theta))^n = r^n(\cos n\theta + j\sin n\theta) = r^n\exp(jn\theta) \text{ for any } n \in \mathbb{Z}$$

(b) For $r = 1$, using the expression given above

$$(\exp(j\theta))^3 = \exp(j3\theta) = \cos 3\theta + j\sin 3\theta$$

On the other hand,

$$\begin{aligned}(\exp(j\theta))^3 &= (\cos\theta + j\sin\theta)^3 \\ &= \cos^3\theta + j3\cos^2\theta\sin\theta - 3\cos\theta\sin^2\theta - j\sin^3\theta \\ &= (\cos^3\theta - 3\cos\theta\sin^2\theta) + j(-\sin^3\theta + 3\cos^2\theta\sin\theta)\end{aligned}$$

By equating real and imaginary parts, we obtain

$$\cos 3\theta = \cos^3\theta - 3\cos\theta\sin^2\theta$$
$$\sin 3\theta = -\sin^3\theta + 3\cos^2\theta\sin\theta$$

(c) *i.* For $n \in \mathbb{Z}^+$ and $k = 0, \pm1, \pm2, \cdots$, $r^n\exp(jn\theta) = \exp(j2\pi k) = \cos(2\pi k) + j\sin(2\pi k) = 1$.

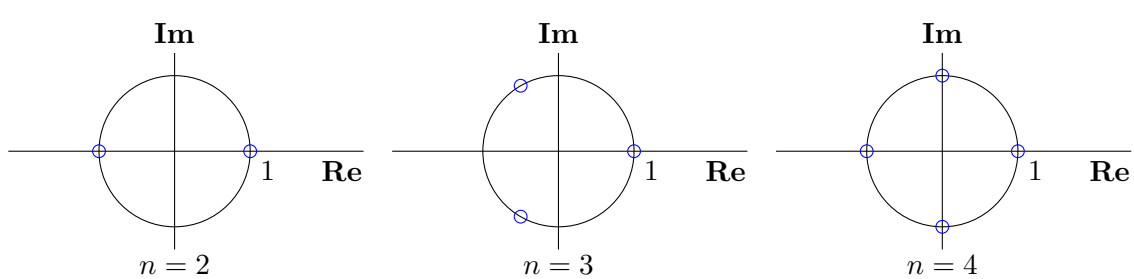
Hence, r^n must have a stable value with nth power changing, and $r = 1$ (remember that $1^a = 1^b$ for any $a, b \in \mathbb{Z}$).

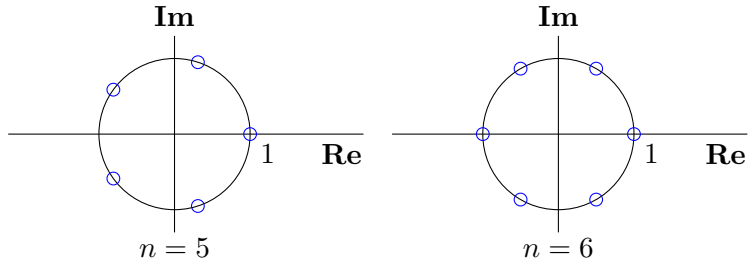
$$r = 1 \implies \exp(jn\theta) = \exp(j2\pi k) \implies \theta = \frac{2\pi k}{n}$$

We know that the complex exponential $\exp(jn\theta)$ is identical to $\exp(jn(\theta + 2\pi))$ for integer n. Hence, we can write the interval of k:

$$\theta = \frac{2\pi k}{n} \text{ for } n \in \mathbb{Z}^+ \text{ and } k = 0, 1, \cdots, n-1$$

ii.





$n = 5$ $\qquad$ $n = 6$

iii. The n-th roots of an arbitrary complex number $w = \rho \exp(j\phi)$ are $\rho^{1/n} \exp\left(j\dfrac{\phi + 2\pi k}{n}\right)$ for $k = 0, 1, \cdots, n-1$.

The cube roots of $27j = 27\exp\left(j\dfrac{\pi}{2}\right)$ are $z_k = 3\exp\left(j\left(\dfrac{\pi}{6} + \dfrac{2\pi}{3}k\right)\right)$ for $k = 0, 1, 2$.

$$z_0 = 3\exp\left(j\frac{\pi}{6}\right) \qquad z_1 = 3\exp\left(j\frac{5\pi}{6}\right) \qquad z_2 = 3\exp\left(j\frac{3\pi}{2}\right)$$

6. (a) Let $S = 1 + z + z^2 + \cdots + z^n$, then

$$S - zS = 1 + z + z^2 + \cdots + z^n - z(1 + z + z^2 + \cdots + z^n) = 1 - z^{n+1}$$

$$S - zS = S(1 - z) = 1 - z^{n+1} \implies S = \frac{1 - z^{n+1}}{1 - z}$$

Then, we obtain the identity

$$1 + z + z^2 + \cdots + z^n = \frac{1 - z^{n+1}}{1 - z} \text{ for any } z \in \mathbb{C}$$

(b) Let $z = r\exp(j\theta)$, where $r = |z|$ and $\theta = \angle z$, then

$$\lim_{n\to\infty} z^{n+1} = \lim_{n\to\infty} |z|^{n+1}(\exp(j\theta))^{n+1}$$

Provided that $|z| < 1$, $|z|^{n+1} \to 0$ as $n \to \infty$. Then, we obtain

$$\lim_{n\to\infty} z^{n+1} = \lim_{n\to\infty} |z|^{n+1}(\exp(j\theta))^{n+1} = 0$$

$$\lim_{n\to\infty} S = \lim_{n\to\infty} \frac{1 - z^{n+1}}{1 - z} = \lim_{n\to\infty} \frac{1}{1 - z} - \underbrace{\lim_{n\to\infty} \frac{z^{n+1}}{1 - z}}_{0} = \frac{1}{1 - z}$$

(c) To derive Lagrange's trigonometric identity

$$1 + \cos\theta + \cos 2\theta + \cdots + \cos n\theta = \frac{1}{2} + \frac{\sin\left[\left(n + \frac{1}{2}\right)\theta\right]}{2\sin\frac{1}{2}\theta}$$

Let $z = \exp(j\theta)$, then use the identity given in part (a)

$$1 + \exp(j\theta) + (\exp(j\theta))^2 + \cdots + (\exp(j\theta))^n = \frac{1 - (\exp(j\theta))^{n+1}}{1 - \exp(j\theta)}$$

By using de Moivre's formula (for $r = 1$) on LHS of expression,

$$1 + (\cos\theta + j\sin\theta) + (\cos 2\theta + j\sin 2\theta) + \cdots + (\cos n\theta + j\sin n\theta) = \frac{1 - (\exp(j\theta))^{n+1}}{1 - \exp(j\theta)}$$

we observe that the LHS of Lagrange's trigonometric identity is identical to the real part of LHS of expression given above.

To obtain the real part of RHS of expression, we perform some algebraic manipulations.

$$\frac{1-(\exp(j\theta))^{n+1}}{1-\exp(j\theta)} = \frac{1-(\exp(j\theta))^{n+1}}{\exp\left(j\frac{\theta}{2}\right)\left[\exp\left(-j\frac{\theta}{2}\right)-\exp\left(j\frac{\theta}{2}\right)\right]} = \frac{-\exp\left(-j\frac{\theta}{2}\right)+(\exp(j\theta))^{n+\frac{1}{2}}}{2j\sin\frac{1}{2}\theta}$$

Then, the real part of RHS of expression is

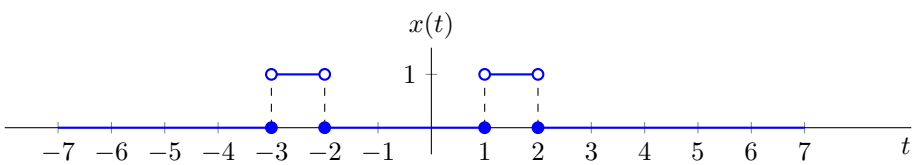
$$\mathbf{Re}\{\frac{1-(\exp(j\theta))^{n+1}}{1-\exp(j\theta)}\} = \frac{1}{2}+\frac{\sin\left[\left(n+\frac{1}{2}\right)\theta\right]}{2\sin\frac{1}{2}\theta}$$

Notice that the sine terms in the numerator belong to real part owing to j term in denominator. Thus, the real part of LHS and RHS of expression give Lagrange's trigonometric identity:

$$1+\cos\theta+\cos 2\theta+\cdots+\cos n\theta = \frac{1}{2}+\frac{\sin\left[\left(n+\frac{1}{2}\right)\theta\right]}{2\sin\frac{1}{2}\theta}$$

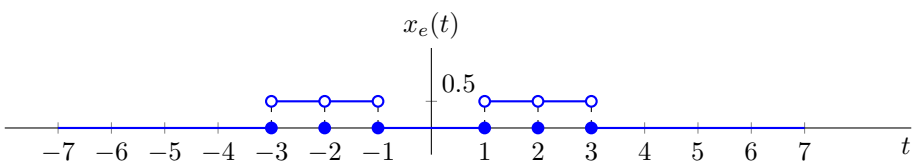
7. (i)

$$x(t) = \begin{cases} 1, & \text{for } -3<t<-2 \text{ or } 1<t<2 \\ 0, & \text{elsewhere} \end{cases}$$

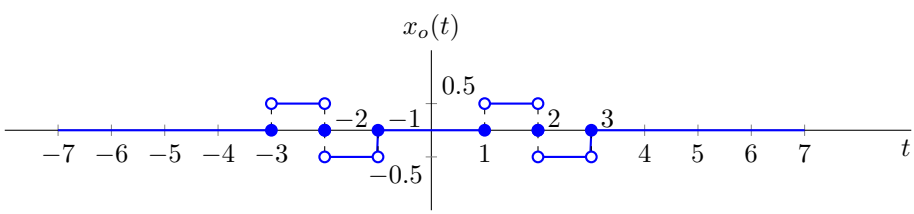


(ii)

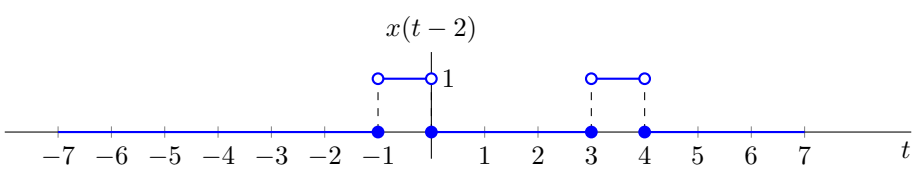
$$x_e(t) = \mathcal{E}\nu(x(t)) = \frac{1}{2}\left(x(t)+x(-t)\right)$$



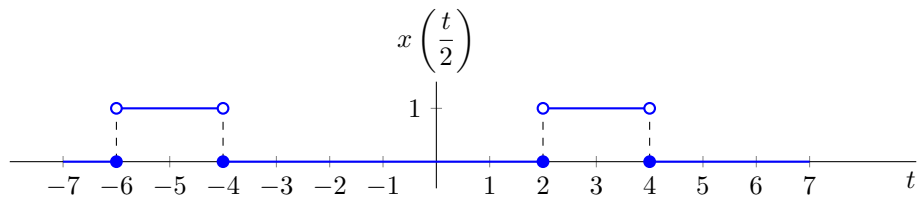
$$x_o(t) = \mathcal{O}d(x(t)) = \frac{1}{2}\left(x(t)-x(-t)\right)$$



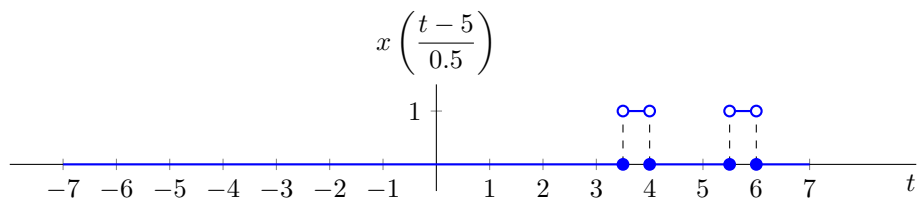
(iii)



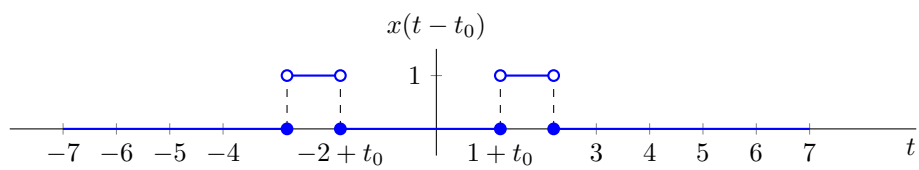
(iv)



(v)



(vi)



To determine the value of t_0 such that $x(t-t_0)$ is an even signal, we can use the fact that the time values at $-2+t_0$ and $1+t_0$ must have opposite signs.Thus, we obtain

$$-2+t_0 = -(1+t_0) \implies t_0 = \frac{1}{2}$$

8. If $x[n]$ is periodic, there exists a positive integer N such that $x[n+N] = x[n]$ and its fundamental period is the smallest nonzero period N satisfying this relation.

(i)

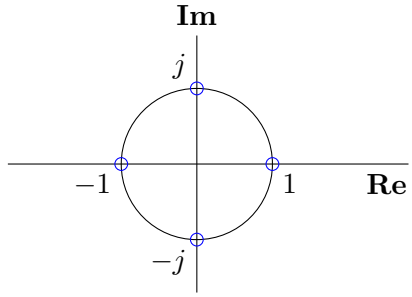
$$x[n] = j^n = \exp\left(j\frac{\pi}{2}n\right)$$

$$x[n+N] = x[n] \implies \exp\left(j\frac{\pi}{2}N\right) = 1 = \exp\left(j2\pi k\right) \text{ for } k \in \mathbb{Z}$$

Then, we obtain $N = 4k$. Hence, the signal is **periodic** and its fundamental period is $N = 4$ (for $k = 1$).

Let $z = 1 = \exp(j\phi)$, then $\phi = 0$ and the 4-th roots of unity on the unit circle (the 4-th roots of z) are $z_k = \exp\left(j\left(\frac{\phi}{4} + \frac{2\pi}{4}k\right)\right) = \exp\left(j\frac{\pi}{2}k\right)$ for $k = 0, 1, 2, 3$.

$z_0 = \exp(j0) = 1 \qquad z_1 = \exp\left(j\frac{\pi}{2}\right) = j \qquad z_2 = \exp\left(j\pi\right) = -1 \qquad z_3 = \exp\left(j\frac{3\pi}{2}\right) = -j$



4-th roots of unity

		It is clear that $x[n]$ visits these roots as n changes: $$x[4m] = 1,\ x[4m+1] = j,\ x[4m+2] = -1 \text{ and } x[4m+3] = -j \text{ for } m \in \mathbb{Z}$$
	(ii)	$$x[n] = z_0^n$$ where the complex constant z_0 is such that $x[n]$ visits 17-th roots of unity as n changes. From the previous part, it is obvious that $$x[n+17] = x[n]$$ and hence, the signal is **periodic** and its fundamental period is $N = 17$ (a prime number). From the periodicity, we know that $$z_0^{17} = 1 = \exp(j2\pi k)$$ and z_0 takes distinct values for $k = 1, 2, \cdots, 16$. Hence, z_0 is not unique.
	(iii)	$$x[n] = \cos n$$ The signal is **not periodic** since there is not any positive integer N satisfying the relation $\cos(n+N) = cos(n)$.
	(iv)	$$x[n] = \left(\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}}j\right)^n = \left(\exp\left(j\frac{\pi}{4}\right)\right)^n = \exp\left(j\frac{\pi}{4}n\right)$$ $$x[n+N] = x[n] \implies \exp\left(j\frac{\pi}{4}N\right) = 1 = \exp\left(j2\pi k\right) \text{ for } k \in \mathbb{Z}$$ Then, we obtain $N = 8k$. Hence, the signal is **periodic** and its fundamental period is $N = 8$ (for $k = 1$).
	(v)	$$x[n] = \mathbf{Re}\left\{\left(\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}}j\right)^n\right\} = \mathbf{Re}\left\{\exp\left(j\frac{\pi}{4}n\right)\right\} = \cos\left(\frac{\pi}{4}n\right)$$ $$\cos\left(\frac{\pi}{4}(n+N)\right) = \cos\left(\frac{\pi}{4}n\right) \implies \frac{\pi}{4}N = 2\pi k \text{ for } k \in \mathbb{Z}$$ Similarly, we obtain $N = 8k$. Hence, the signal is **periodic** and its fundamental period is $N = 8$ (for $k = 1$).
	(vi)	$$x[n] = \exp\left(j\frac{25}{4}\pi n\right)$$ $$x[n+N] = x[n] \implies \exp\left(j\frac{25}{4}\pi N\right) = 1 = \exp\left(j2\pi k\right) \text{ for } k \in \mathbb{Z}$$ Then, we obtain $N = \frac{8}{25}k$. Hence, the signal is **periodic** and its fundamental period is $N = 8$ (for $k = 25$). Actually, $x[n]$ is identical to the signal given in part (iv). $$x[n] = \exp\left(j\frac{25}{4}\pi n\right) = \exp\left(j\left(6 + \frac{1}{4}\right)\pi n\right) = \underbrace{\exp\left(j6\pi n\right)}_{1}\exp\left(j\frac{\pi}{4}n\right) = \exp\left(j\frac{\pi}{4}n\right)$$ Therefore we obtain the same results.